DSA0601 DATA HANDLING and VISUALISATION

ACTIVITY - 5

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1. Geospatial Data Visualization

Dataset

City	Latitude	Longitude	Population (Millions)
Chennai	13.08	80.27	11
Bangalore	12.97	77.59	13
Hyderabad	17.38	78.48	10
Mumbai	19.07	72.87	20
Delhi	28.61	77.21	32
Pune	18.52	73.85	7
Kolkata	22.57	88.36	15
Ahmedabad	23.02	72.57	8
Jaipur	26.91	75.79	4
Kochi	9.93	76.26	2

CODE:

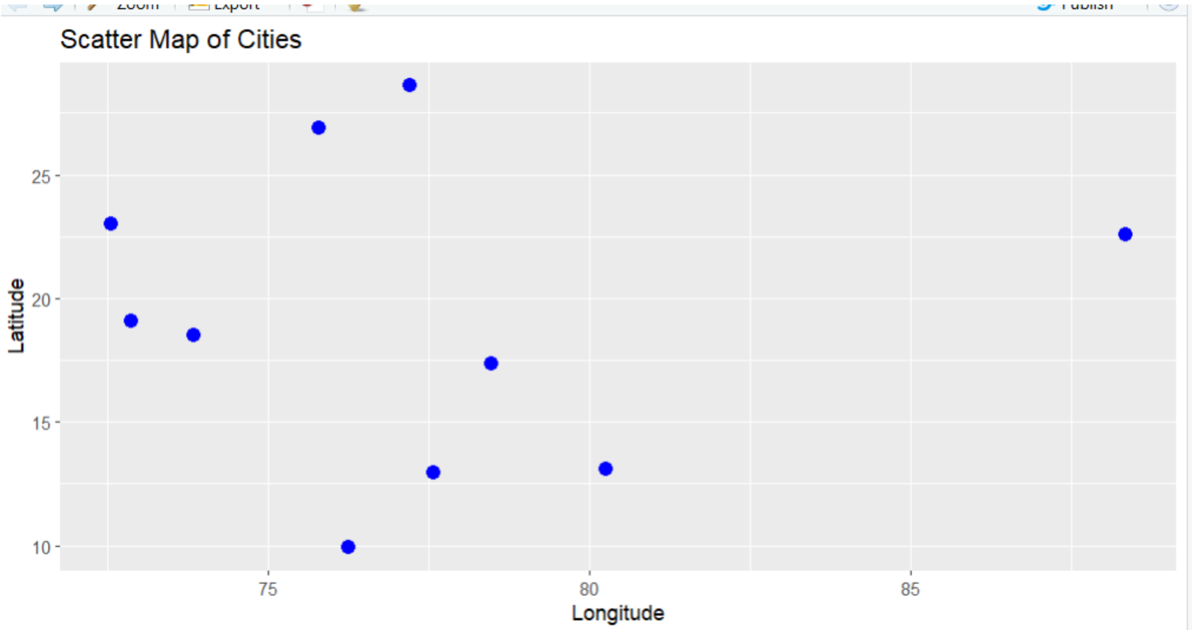
```
# Load library
library(ggplot2)
# Dataset
city <- data.frame(
  City = c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi",
```

```r
        "Pune","Kolkata","Ahmedabad","Jaipur","Kochi"),
  Latitude =
c(13.08,12.97,17.38,19.07,28.61,18.52,22.57,23.02,26.91,9.93),
  Longitude =
c(80.27,77.59,78.48,72.87,77.21,73.85,88.36,72.57,75.79,76.26),
  Population = c(11,13,10,20,32,7,15,8,4,2)
)
# Q1 Scatter Map
ggplot(city, aes(x=Longitude, y=Latitude)) +
  geom_point(size=3, color="blue") +
  ggtitle("Scatter Map of Cities")
# Q2 Bubble Map
ggplot(city, aes(Longitude, Latitude, size=Population)) +
  geom_point(color="red") +
  ggtitle("Bubble Map")
# Q3 Labels
ggplot(city, aes(Longitude, Latitude, size=Population)) +
  geom_point(color="blue") +
  geom_text(aes(label=City), vjust=-1)
# Q4 Colors based on Population
ggplot(city, aes(Longitude, Latitude)) +
  geom_point(aes(size=Population, color=Population)) +
  geom_text(aes(label=City), vjust=-1)
# Q5 Highest Population
```

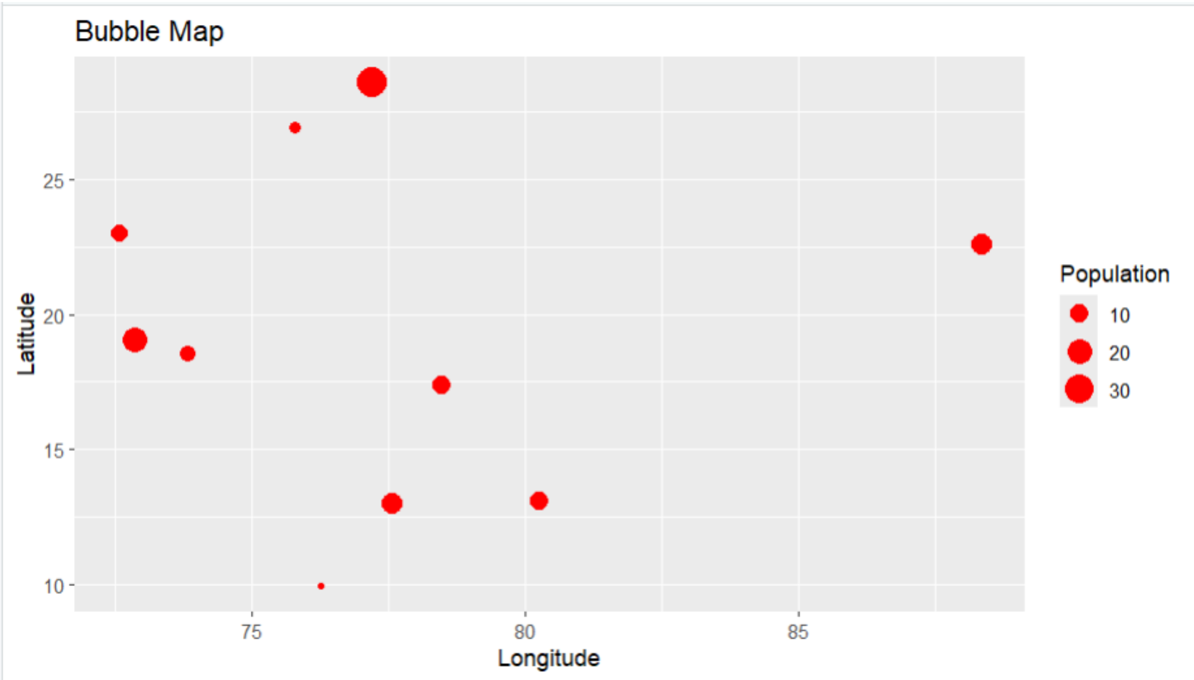
city[which.max(city$Population),]

Questions

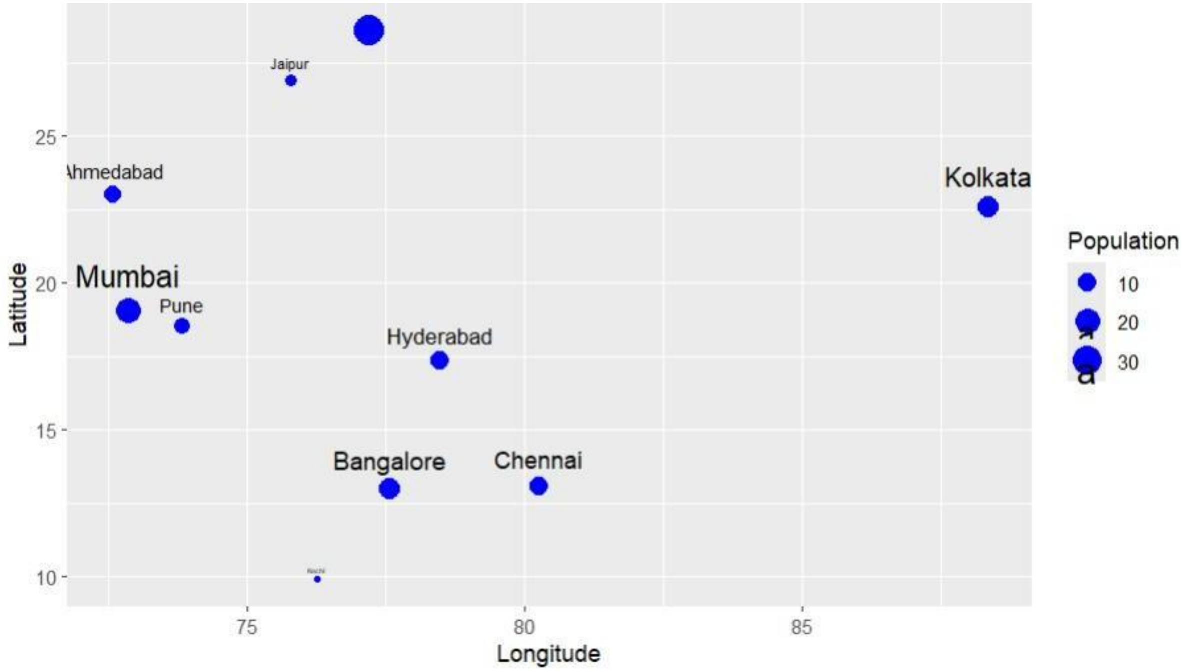
Q1. Create a scatter map showing all cities using longitude and latitude.



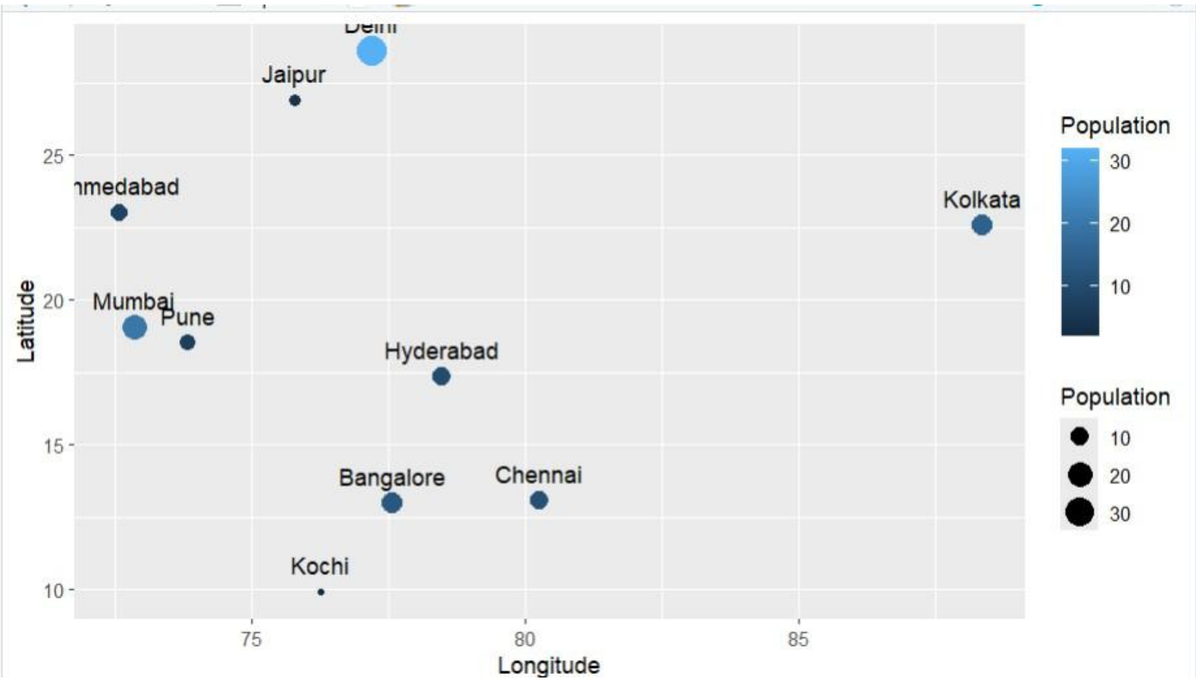
Q2. Plot a bubble map where bubble size represents population.



Q3. Label each city on the plot using geom_text().



Q4. Use different colors based on population size.



Q5. Identify the city with the highest population from the visualization.

```
> city[which.max(city$Population), ]
   City Latitude Longitude Population
5 Delhi    28.61     77.21         32
```

2. Network Visualization

Dataset

From	To
A	B
A	C
B	D
C	D
C	E
D	E
E	F
F	G
G	H
H	A

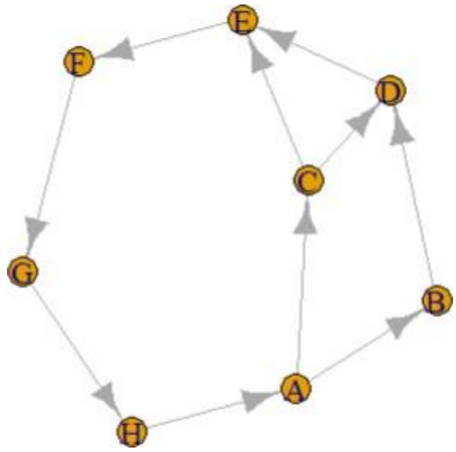
CODE

```
library(igraph)

# Dataset

edges <- data.frame(

  From=c("A","A","B","C","C","D","E","F","G","H"),

  To=c("B","C","D","D","E","E","F","G","H","A")

)

g <- graph_from_data_frame(edges)

# Q1 Network Graph

plot(g)
```

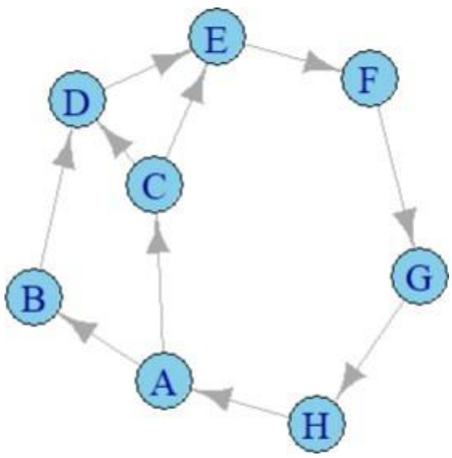
```
# Q2 Labels and Colors
plot(g,
     vertex.color="skyblue",
     vertex.size=30,
     vertex.label.cex=1.2)
# Q3 Degree
degree(g)
# Q4 Highest Degree Node
deg <- degree(g)
deg
names(deg[deg==max(deg)])
# Q5 Circular Layout
plot(g,
     layout=layout_in_circle(g),
     vertex.color="orange",
     vertex.size=30)
```

Questions

Q1. Create a network graph using the igraph package.



Q2. Display node labels and customize node colors.



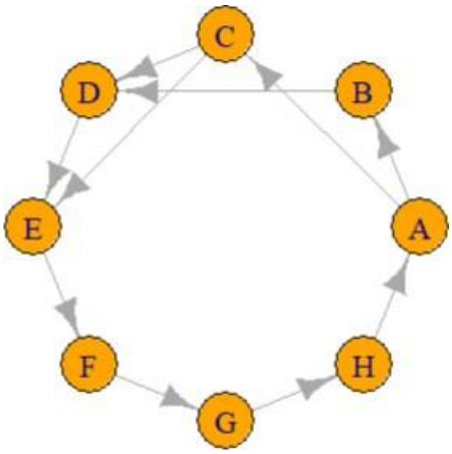
Q3. Calculate the degree of each node.

```
> # Q3 Degree
> degree(g)
A B C D E F G H
3 2 3 3 3 2 2 2
```

Q4. Identify the node with the highest number of connections.

```
> # Q4 Highest Degree Node
> deg <- degree(g)
> deg
A B C D E F G H
3 2 3 3 3 2 2 2
```

Q5. Plot the network using a circular layout.



3. Temporal Data Representation

Dataset

Month	Sales (₹ Lakhs)
Jan	120
Feb	135
Mar	150
Apr	170
May	190
Jun	220
Jul	210
Aug	230
Sep	250

Oct	270
Nov	290
Dec	320

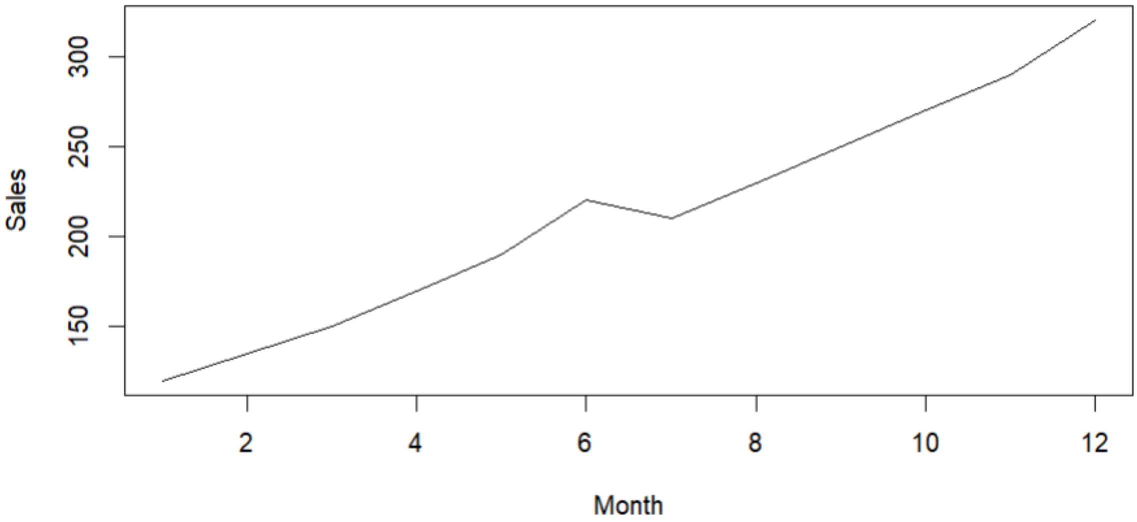
CODE

```
# Dataset
months <- c("Jan","Feb","Mar","Apr","May","Jun",
           "Jul","Aug","Sep","Oct","Nov","Dec")
sales <- c(120,135,150,170,190,220,210,230,250,270,290,320)
# Q1 Line Chart
plot(sales,
     type="l",
     xlab="Month",
     ylab="Sales")
# Q2 Month Names
plot(sales,
     type="l",
     xaxt="n",
     xlab="Month",
     ylab="Sales")
axis(1, at=1:12, labels=months)
# Q3 Highlight Points
```

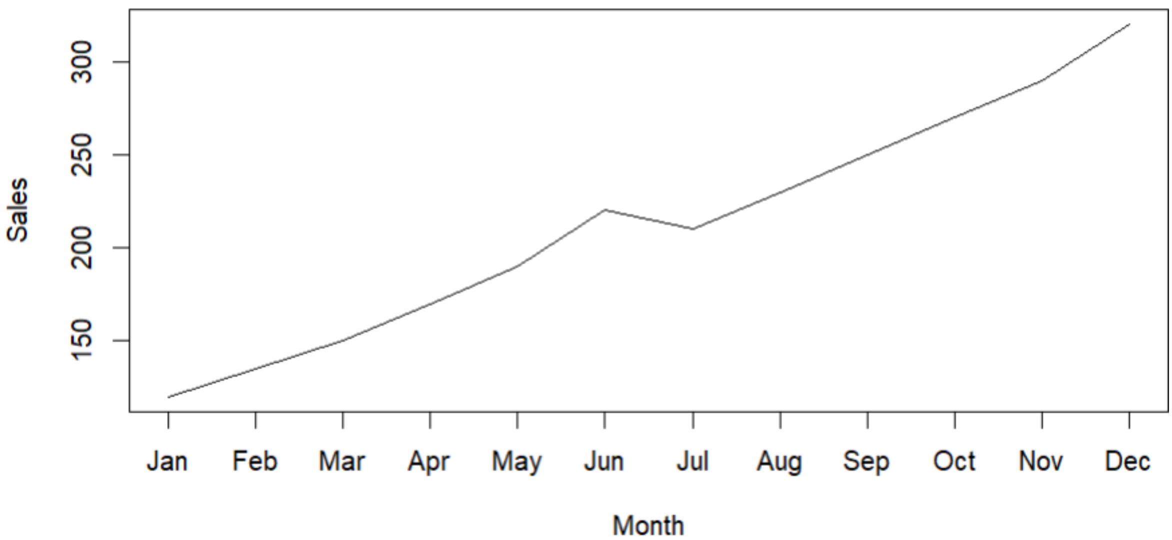
```
plot(sales,
    type="o",
    pch=19,
    xaxt="n")
axis(1, at=1:12, labels=months)
# Q4 Grid Lines
plot(sales,
    type="o",
    pch=19,
    xaxt="n",
    xlab="Month",
    ylab="Sales")
axis(1, at=1:12, labels=months)
grid()
# Q5 Maximum Sales
months[which.max(sales)]
max(sales)
```

Questions

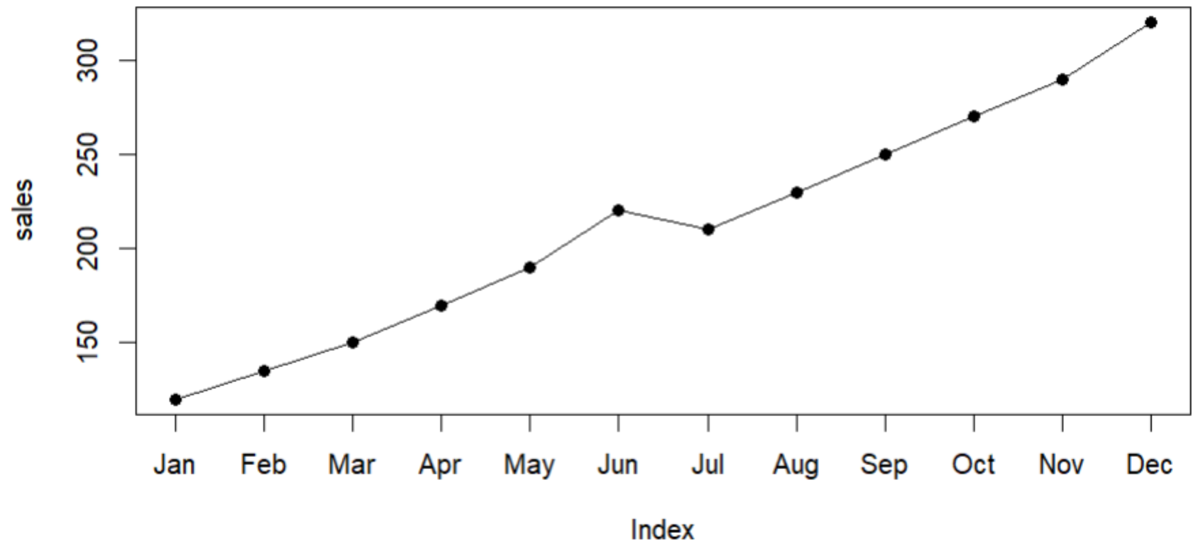
Q1. Create a line chart showing monthly sales.



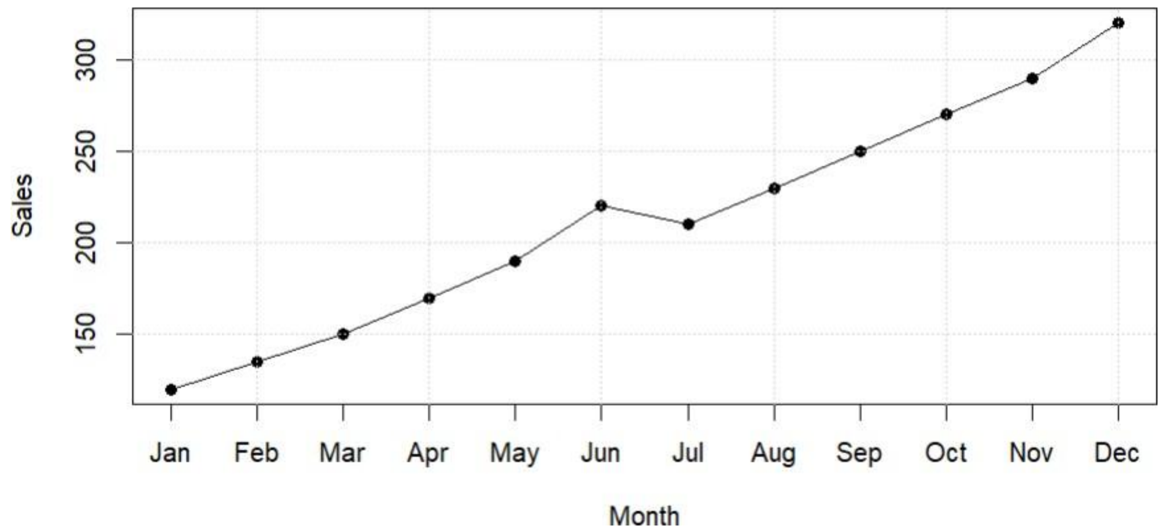
Q2. Add custom month names using the axis() function.



Q3. Highlight data points using pch=19.



Q4. Display grid lines and proper axis labels.



Q5. Identify the month with the maximum sales from the graph.

```
> # Q5 Maximum Sales
> months[which.max(sales)]
[1] "Dec"
> max(sales)
[1] 320
```

4. Multivariate Data Visualization

Dataset

Student	Marks	Attendance (%)	Study Hours	Projects
A	60	75	2	1
B	68	80	3	2
C	74	84	4	2
D	79	88	5	3
E	82	90	5	3
F	85	92	6	4
G	88	94	6	4
H	91	95	7	5
I	94	97	8	5
J	97	99	9	6

CODE

```
# Dataset
student <- data.frame(
  Student=c("A","B","C","D","E","F","G","H","I","J"),
  Marks=c(60,68,74,79,82,85,88,91,94,97),
  Attendance=c(75,80,84,88,90,92,94,95,97,99),
  StudyHours=c(2,3,4,5,5,6,6,7,8,9),
  Projects=c(1,2,2,3,3,4,4,5,5,6)
)
# Q1 Bubble Plot
symbols(student$Marks,
```

student$Attendance,

circles=student$StudyHours,

inches=0.3,

bg="blue",

xlab="Marks",

ylab="Attendance")

Q2 Scatter Plot Matrix

pairs(student[,2:5])

Q3 Correlation

cor(student[,2:5])

Q4 Highest Marks Student

student[which.max(student$Marks),]

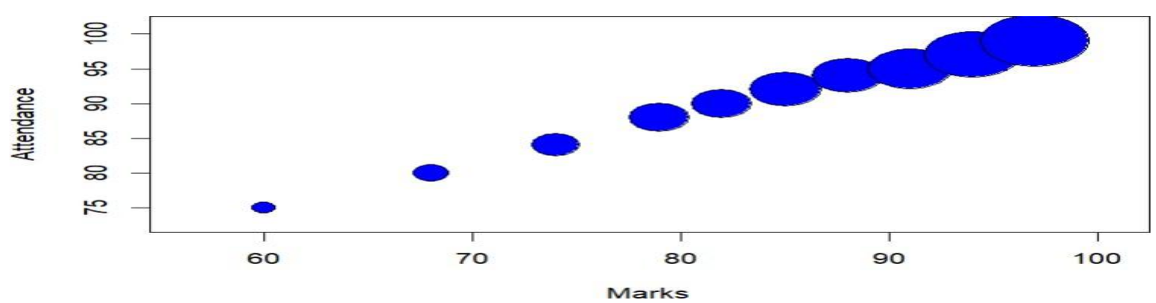
Q5 Heatmap

heatmap(cor(student[,2:5]), symm=TRUE)

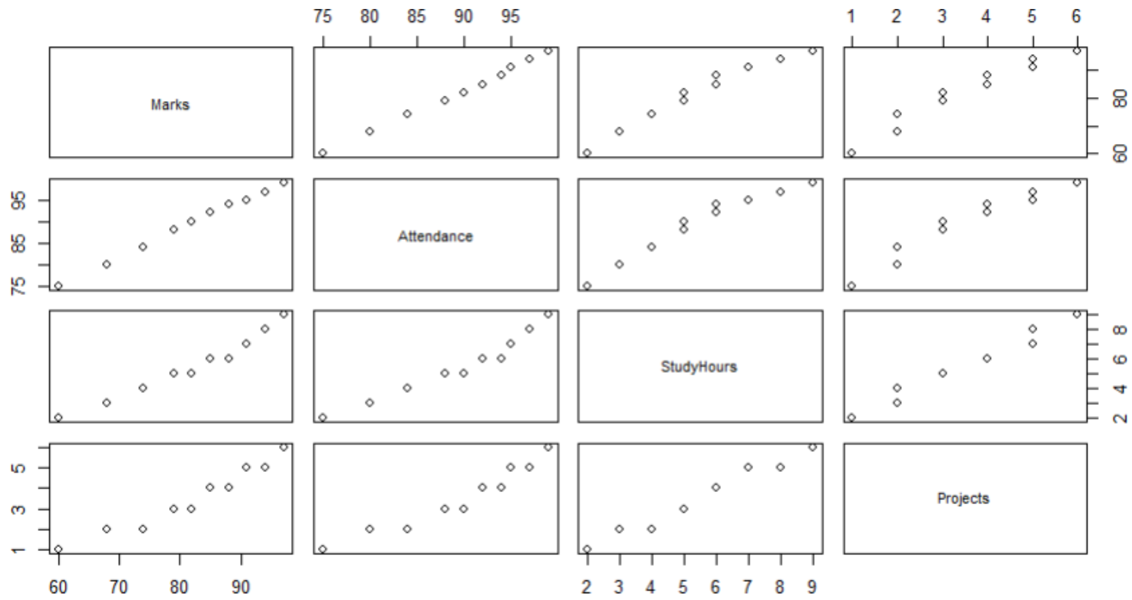
Questions

Q1. Create a bubble plot where:

- X-axis = Marks
- Y-axis = Attendance
- Bubble size = Study Hours



Q2. Produce a scatter plot matrix using the pairs() function.



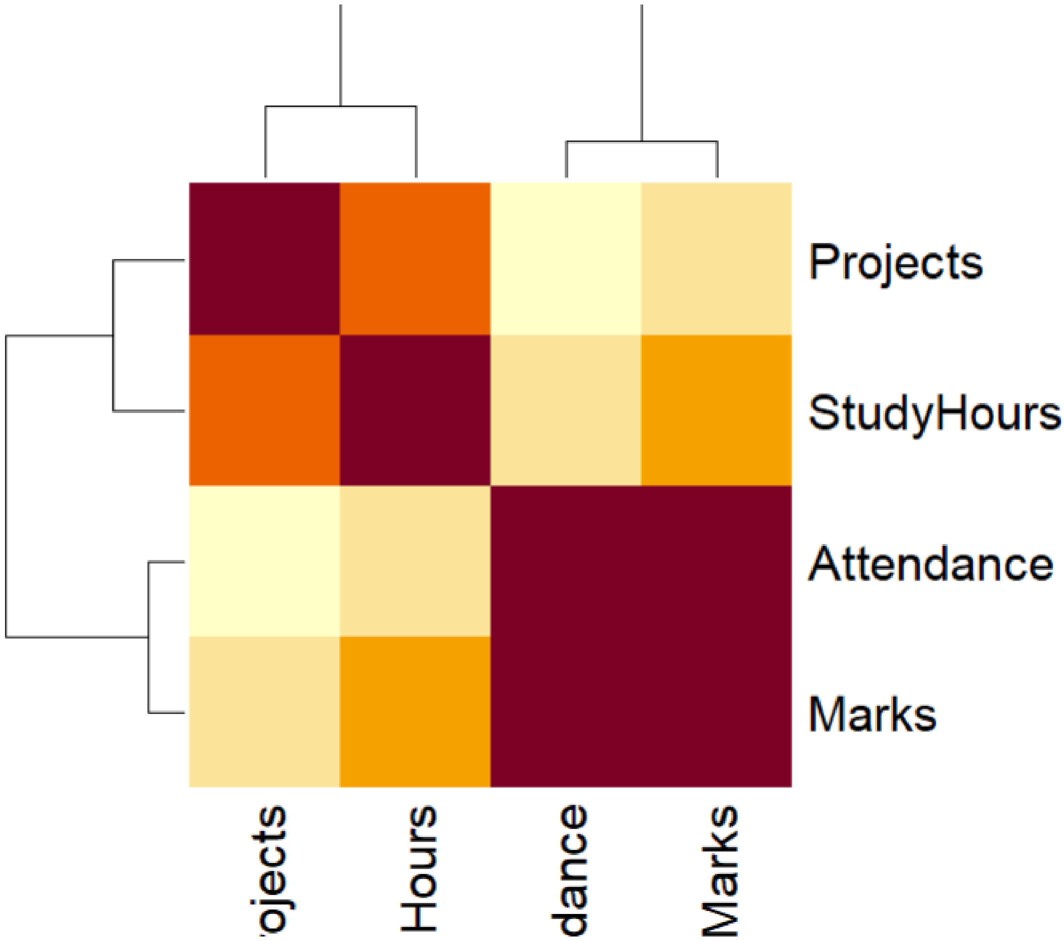
Q3. Find the correlation among Marks, Attendance, and Study Hours.

```
> # Q3 Correlation
> cor(student[,2:5])
               Marks Attendance StudyHours
Marks      1.0000000  0.9984669  0.9814683
Attendance 0.9984669  1.0000000  0.9735953
StudyHours 0.9814683  0.9735953  1.0000000
Projects   0.9728786  0.9648732  0.9863115
            Projects
Marks      0.9728786
Attendance 0.9648732
StudyHours 0.9863115
Projects   1.0000000
```

Q4. Identify the student with the highest marks and largest bubble.

```
> # Q4 Highest Marks Student
> student[which.max(student$Marks), ]
   Student Marks Attendance StudyHours
10       J    97         99          9
   Projects
10        6
```

Q5. Create a heatmap showing the relationships among all numerical variables.



5. Integrated Data Visualization Exercise

Dataset

Year	Sales	Profit	Customers	Branches
2016	150	20	1200	5
2017	170	25	1500	6
2018	200	32	1800	7
2019	240	40	2200	8
2020	260	42	2400	8
2021	300	55	2800	10
2022	340	65	3300	11
2023	390	72	3800	12
2024	430	82	4200	13
2025	480	95	4700	15

CODE

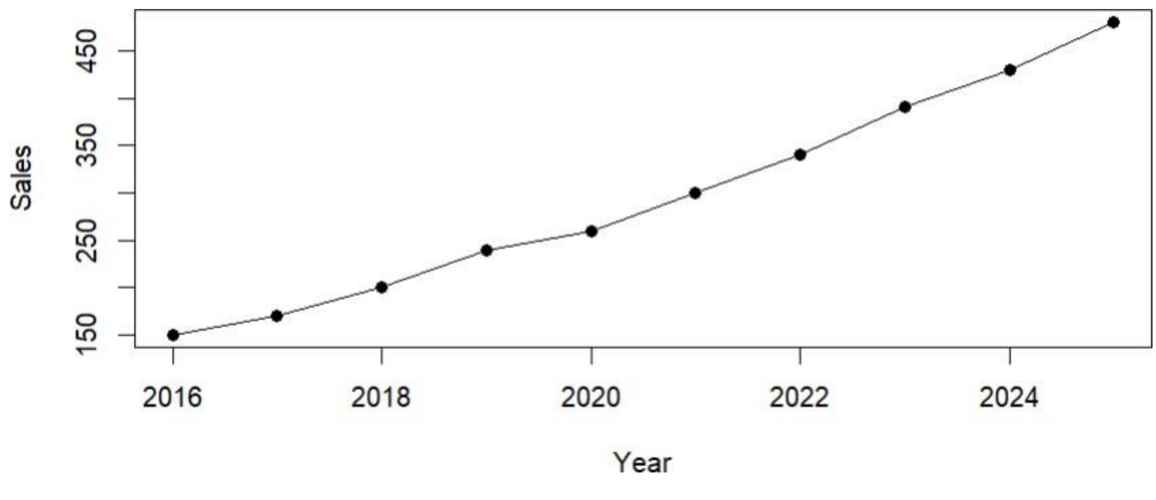
```
# Dataset

business <- data.frame(

  Year=2016:2025,

  Sales=c(150,170,200,240,260,300,340,390,430,480),

  Profit=c(20,25,32,40,42,55,65,72,82,95),


Customers=c(1200,1500,1800,2200,2400,2800,3300,3800,4200,4700)
,

  Branches=c(5,6,7,8,8,10,11,12,13,15)

)

# Q1 Line Chart

plot(business$Year,
```

```r
    business$Sales,
    type="o",
    pch=19,
    xlab="Year",
    ylab="Sales")
# Q2 Bubble Plot
symbols(business$Sales,
        business$Profit,
        circles=business$Customers,
        inches=0.3,
        bg="green",
        xlab="Sales",
        ylab="Profit")
# Q3 Correlation Matrix
cor(business[,2:5])
# Q4 Scatter Plot Matrix
pairs(business[,2:5])
# Q5 Branches vs Sales & Profit
plot(business$Branches,
    business$Sales,
    pch=19,
    col="blue",
    xlab="Branches",
    ylab="Sales",
```

```
    main="Branches vs Sales")
plot(business$Branches,
    business$Profit,
    pch=19,
    col="red",
    xlab="Branches",
    ylab="Profit",
    main="Branches vs Profit")

cor(business[,2:5])
```

Questions

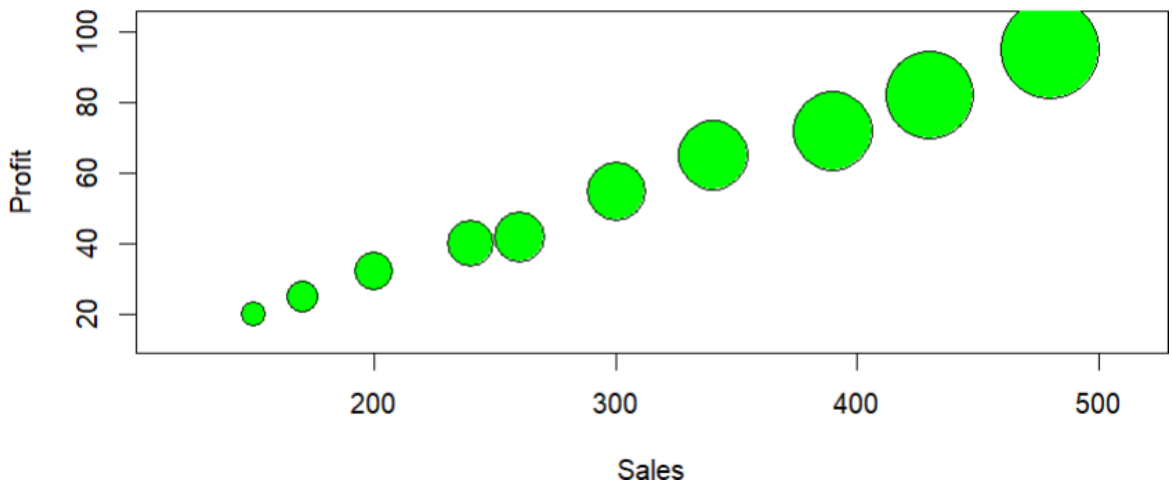
Q1. Create a temporal line chart showing yearly sales.



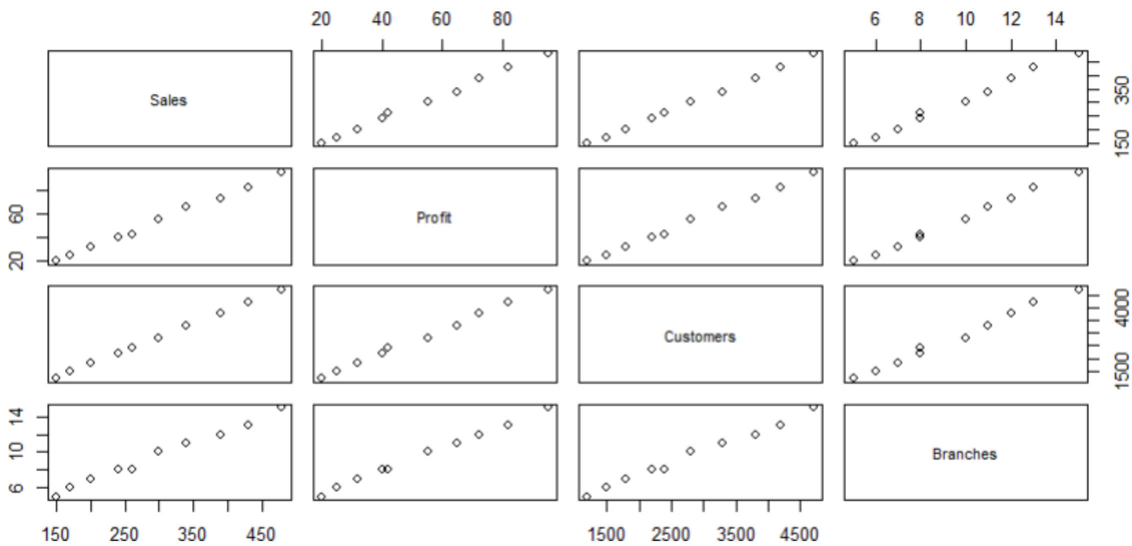
Q2. Generate a multivariate bubble plot with:

- X-axis = Sales
- Y-axis = Profit

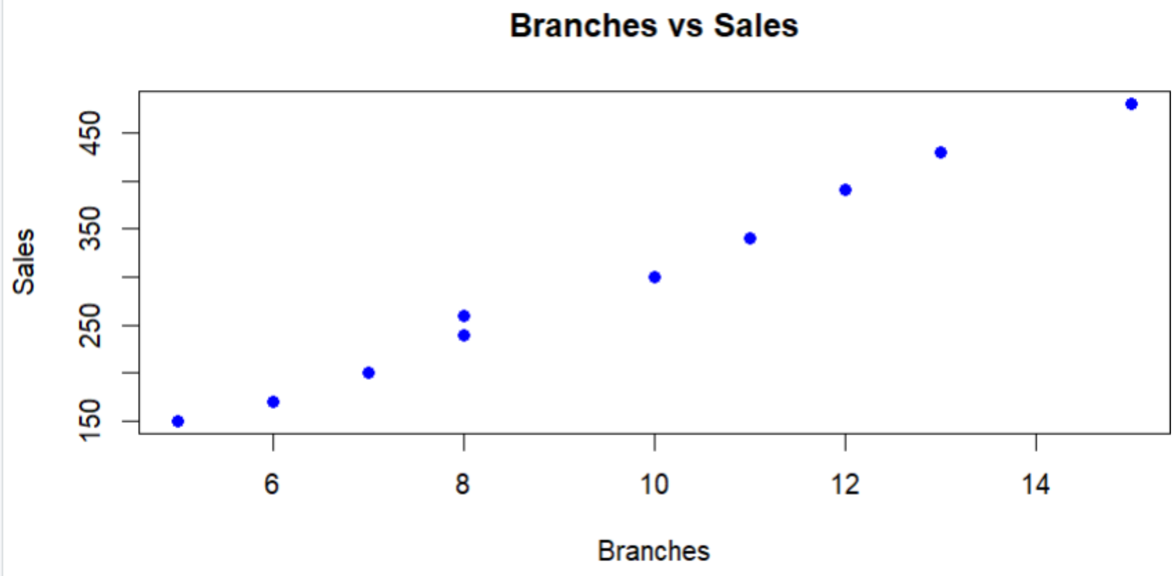
 Bubble size = Customers



Q3. Compute the correlation matrix for Sales, Profit, Customers, and Branches.



Q4. Create a scatter plot matrix using the pairs() function.



Q5. Interpret the visualizations to determine whether an increase in branches is associated with higher sales and profits.